

Analyzing Crop Yield Trends and Climate Impact in the United States

1. Research Question & Motivation

Agriculture plays a critical role in both the economy and food supply of the United States. Understanding trends in crop yield and production across regions and over time is essential for evaluating agricultural efficiency, sustainability, and long-term food security. These trends are influenced by a combination of factors, including technological advancements, farming practices, and environmental conditions. Our research question is: What are the key trends in crop yield and production across U.S. states, and how can climate conditions during the growing season help explain these patterns?

To enable a more focused and in-depth analysis, this study narrows its scope to a single crop. Corn was selected as the primary crop of focus due to its widespread production, consistent data availability, and economic importance within the U.S. agricultural system. Additional crops are included to provide context and allow for broader comparisons.

The goal of this project is first to identify patterns in crop yield and production, and then to analyze how climate factors, particularly temperature and precipitation, help explain these observed trends. The analysis and accompanying interactive dashboard are designed for a non-technical audience, such as policymakers and economic researchers, who require clear, high-level insights into agricultural performance rather than detailed statistical modeling.

2. Data & Methodology

This project combines three main datasets: crop yield data, corn production and pricing data, and monthly weather data containing temperature and precipitation measurements. The crop yield data was sourced from USDA NASS quick stats website. The corn production data was accessed from Kaggle which was sourced from the USDA NASS quick stats. Additionally, the weather data was accessed from the National Centers for Environmental Information. To ensure consistency and comparability, several preprocessing steps were applied. Yield values were standardized into pounds per acre so that different crops could be compared on a common scale. Missing values and formatting inconsistencies were cleaned to improve overall data quality.

Weather data was filtered to focus on the growing season of corn, defined as April through September, and aggregated at the state and year level. These aggregated climate variables were then merged with crop yield data, allowing for direct comparison between environmental conditions and agricultural output. The processed dataset was used to build an interactive Streamlit dashboard, where users can explore geographic distributions of crop yield, analyze trends over time, examine production and revenue patterns, and investigate how climate variables relate to crop performance. Additional visualizations were designed to help identify optimal climate ranges and classify regions based on their suitability for crop growth.

3. Key Findings

The analysis reveals clear patterns in crop yield trends across the United States. Crop yields vary significantly by region, with Midwestern states consistently demonstrating higher productivity.

These regions benefit from a combination of favorable environmental conditions and well-developed agricultural infrastructure. Over time, yields have generally increased, reflecting improvements in farming technology, irrigation systems, and crop management practices.

Further analysis shows that climate conditions help explain many of these observed patterns. Moderate temperature ranges are associated with higher yields, while extreme heat or cold tends to reduce productivity. Precipitation follows a similar pattern, where both insufficient and excessive rainfall negatively affect crop output. By examining the relationship between temperature, precipitation, and yield, a clear “optimal zone” emerges. For corn, this range is typically around 63°F to 69°F in average temperature and 28 to 34 inches of total precipitation during the growing season.

Using these thresholds, states were categorized into optimal, semi-optimal, and non-optimal climate conditions. States within the optimal range generally achieve higher yields, supporting the idea that climate plays a major role in agricultural performance. However, some states perform well even outside these conditions, indicating that other factors such as irrigation, soil quality, and technological advancements can offset less favorable climates.

Another important distinction highlighted in the analysis is the difference between yield and production. Yield measures efficiency per acre, while production reflects total output. Some states achieve high production not because of higher efficiency, but because they cultivate larger areas of land. This distinction is important for understanding agricultural performance from both an efficiency and scale perspective.

4. Assumptions and Limitations

There are several limitations that should be considered when interpreting these results. Weather data is aggregated at the state level, which can hide local variations such as regional droughts or flooding. The growing season was simplified into a fixed range of months rather than using crop-specific growth cycles, which may not fully capture real-world conditions.

Additionally, yield standardization relies on approximate conversion factors, which may introduce small inaccuracies. The analysis also does not include other important variables such as soil quality, fertilizer use, pest control, or technological differences across regions. Because of these limitations, the results should be interpreted as showing general relationships rather than direct cause-and-effect conclusions.

5. Conclusion

This project identifies clear trends in crop yield and production across the United States, highlighting consistent regional differences and long-term improvements in productivity. Midwestern states emerge as key contributors, reflecting both favorable conditions and strong agricultural systems.

Climate analysis helps explain these trends by showing that moderate temperature and precipitation ranges are associated with higher yields. However, climate alone does not fully determine productivity, as technological advancements and farming practices also play a significant role.

The interactive dashboard developed in this project provides a practical tool for exploring both yield trends and the factors influencing them. By combining trend analysis with environmental context, the project offers a more complete understanding of how agricultural performance evolves over time and across regions.